

HUMAN RESOURCE WORKFORCE ANALYSIS AND EMPLOYEE PERFORMANCE INSIGHTS

Project Overview

This project presents a comprehensive Human Resources (HR) analytics solution built using Power BI. It consolidates employee-related data into two interactive dashboards — **HR Overview Dashboard** and **HR Analysis Dashboard** — enabling HR managers and business leaders to monitor workforce performance, compensation, attrition, training, and demographic trends.

This project demonstrates how data visualization can support strategic HR decision-making and improve organizational efficiency.

Project Objectives

- Deliver a unified platform for all **HR reporting** needs with real-time insights.
- Analyse **employee demographics**, job roles, gender distribution, and work shifts.
- Evaluate **income patterns**, performance scores, and experience levels.
- Identify **attrition drivers** using advanced analytics (Key Influencers).
- Assess **promotion eligibility** and training distribution across roles.
- Support HR leaders in **workforce planning**, talent management, and policy development.
- Replace manual reporting with **automated, interactive Power BI dashboards**.

Data Sources

- **Source Description and Timeline:** UCI Machine Learning Repository/ Google Dataset Search (2020-2025).
- **Domain:** Human Resource

Problem Statement

- Build a **centralized, visual platform** that consolidates all major HR metrics for quick and accurate decision-making.
- Enable HR teams to **proactively identify talent gaps** and strengthen workforce planning using clear visual insights.

- Support leadership in **recognizing high-performing departments and roles**, encouraging targeted development and reward strategies.
- Provide a unified platform to **monitor employee well-being indicators** such as leave patterns, training hours, and work shifts.
- Help the organization **reduce attrition** by uncovering key influencers and patterns behind employee turnover.
- Improve transparency by offering **easy-to-interpret metrics** on income distribution, performance scores, and promotion eligibility.
- Empower decision-makers with **real-time analytics** that enhance policy formulation and operational efficiency.
- Facilitate a culture of **continuous improvement** by highlighting areas where training, experience, or departmental support can be strengthened.
- Promote **data-driven HR strategies** that align workforce performance with organizational goals and long-term growth.

Attribute Details

Attribute Name	Data Type	Description
Employee ID	Integer	Unique identifier assigned to each employee.
Age	Integer	Employee's age in completed years.
Department	String	Department where the employee works (e.g., HR, Sales).
Job Role	String	Specific job position or designation of the employee.
Gender	String	Gender identity of the employee (Male, Female, Others).
Monthly Income	Float	Employee's monthly salary or earnings.
Work Experience	Integer	Total years of professional experience.
Education Level	String	Highest educational qualification (e.g., Bachelor, Master, Doctorate).
Training Hours	Float	Total number of training hours completed by the employee.

Performance Score	Float	Employee's performance rating on a defined scale (e.g., 1–5).
Leave Days Taken	Integer	Number of leave days taken within a specific period.
Attendance Rate Pct	Float	Percentage of attendance maintained by the employee.
Attrition	String	Indicates whether the employee has left the organization.
Work Shift	String	Assigned work shift (Day, Night, Rotational, Flexible).
Promotion Eligibility	String	Indicates whether the employee qualifies for promotion based on criteria.
Income Summary	String	Categorized income level (Low, Medium, High) derived from Monthly Income.
Experience Category	String	Grouped experience level (Junior, Mid, Senior) derived from Work Experience.
Attendance Policy	String	Classification of attendance behavior (e.g., Good, Average, Poor) based on attendance metrics.

Tools & Technologies

1. Microsoft Excel

- Microsoft Excel was used to clean, structure, and prepare the raw HR dataset before importing it into Power BI.
- Key tasks included removing duplicates, fixing missing or inconsistent values, formatting columns, creating calculated fields, and performing quick exploratory checks. Excel also supported initial validation to ensure data accuracy.
- Overall, it served as the foundation for producing a clean, reliable dataset ready for advanced analysis and visualization in Power BI.

2. Microsoft Power BI

- Power BI was used to convert the prepared HR dataset into interactive dashboards and advanced analytical visuals.
- It supported data modelling through relationships, DAX measures, and calculated columns, enabling deeper workforce analysis. The dashboards included KPIs, charts, tree maps, scatter plots, Key Influencers, and decomposition trees to uncover attrition drivers and departmental trends.
- Automated data refresh ensured real-time updates, while intuitive visual storytelling helped HR leaders interpret workforce performance, income patterns, and attrition insights effectively.
- Overall, Power BI served as the core tool for delivering meaningful, interactive HR analytics.

Data Pre-Processing (Excel)

Tasks Performed:

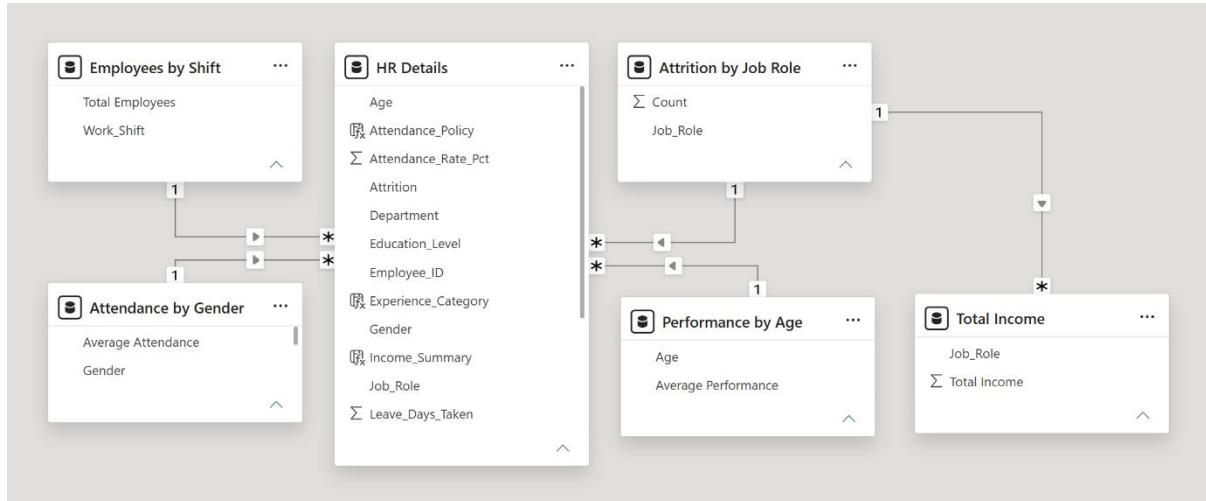
- **Removed duplicate records** to ensure each employee entry was unique.
- **Handled missing values** in key fields such as age, work experience, Attrition, shift, gender and Leave days taken.
- **Corrected inconsistent text entries** (e.g., gender labels, department names, job roles).
- **Standardized data types** by converting columns like Age, Monthly Income, Work Experience, and Performance Score into numeric formats.
- **Used Conditional Formatting to highlight** or visually distinguish cells that meet specific criteria. This makes it easy to spot trends, identify outliers, and quickly understand the data at a glance.

Data Modelling and DAX (Power BI)

Data Modelling:

- A structured data model was created in Power BI by establishing relationships between multiple tables. The **HR Details** table functions as the **fact table**, containing core employee metrics such as income, performance, attendance, experience, and attrition.
- Supporting tables including **Employees by Shift**, **Attrition by Job Role**, **Attendance by Gender**, **Performance by Age**, and **Total Income** were designed as **dimension tables**. These dimension tables provide descriptive attributes that enrich the analysis.

- By connecting the fact table with its corresponding dimension tables, the model achieves a **Star Schema** structure, ensuring clear cardinality, optimized performance, and efficient filtering across all HR dashboards.



Calculated Columns & DAX Measures:

Using DAX (Data Analysis Expressions) many calculated columns and measures are created to enhance analytical insights.

- Calculated Columns:**
 - Promotion Eligibility
 - Income Summary
 - Experience Category
 - Attendance Policy
- Calculated Measures:**
 - Employee Count
 - Total Income
 - Average Income
 - High Performance Income
 - Attrition Count
 - Average Training Hours
 - Average Leave

Analysis and Visualizations (Power BI)

1. HR Overview Dashboard

The HR Overview Dashboard provides a consolidated snapshot of the organization's workforce, enabling quick understanding of employee distribution, income patterns, performance trends, and demographic insights.

Key Analytical Findings

- **Workforce Composition:** The organization has a total of **500 employees**, with a balanced gender distribution across job roles. This indicates healthy diversity and representation.
- **Income Insights:**
 - Total income amounts to **₹3.83M**, with an average monthly income of **₹7.66K**.
 - The **Research & Development** department contributes the highest income, followed by **Sales** and **Human Resources**.
 - High-performing employees contribute significantly to overall income, highlighting the impact of performance on compensation.
- **Performance Trends:** Performance scores show a steady increase with age, suggesting that experience positively influences employee output.
- **Promotion Eligibility:** Promotion eligibility is strongly linked to income categories, with medium-income employees forming the largest eligible group. This helps identify potential talent pipelines.
- **Training & Leave Patterns:** Average training hours (32.98) and leave days (11.69) indicate balanced employee development and work-life integration.

Visualizations Included

- **Ranking:** Department wise Total Income
- **Gauge Chart:** Average Income
- **Stacked Column Chart:** Count of job role by gender
- **Line Chart:** Performance Score by Age
- **Area Chart:** Work Shift vs Work Experience
- **Donut Chart:** Employee Count by Promotion Eligibility
- **Pie Chart:** Monthly Income by Job Role
- **Decomposition Tree:** Attrition by Education Level → Department

2. HR Analysis Dashboard

The HR Analysis Dashboard dives deeper into departmental performance, attrition patterns, experience levels, and income distribution, enabling strategic workforce planning.

Key Analytical Findings

- **Departmental Performance:** Employees with Bachelor's and Master's degrees show higher performance scores, indicating strong academic influence on productivity.
- **Attendance Trends:** Attendance rates vary across education levels, helping HR identify groups that may require engagement or policy adjustments.
- **Attrition Insights:**
 - Total attrition count is **87**, with the highest attrition in **Research & Development**, followed by **Sales**.
 - This highlights departments where retention strategies may be needed.
- **Experience vs Income:** High-income employees show significantly higher work experience, confirming a positive correlation between experience and compensation.
- **Job Role Income Distribution:** Managerial and director-level roles contribute the highest income, while entry-level roles show moderate income distribution.
- **Key Influencers:** Attrition is influenced by factors such as age (≤ 21), low work experience (≤ 20), and specific job roles like Healthcare Representative. These insights help HR identify at-risk employee groups.

Visualizations Included

- **Ribbon Chart:** Sum of Attendance rate by Education level
- **Combo Chart:** Sum of Performance Score by Education Level
- **Waterfall Chart:** Attrition Count by Department
- **Funnel Chart:** Sum of work experience by income summary
- **Scatter Plot:** Monthly Income vs Performance Score
- **Tree map:** Training Hours by Job Role
- **Table:** Income Summary by Experience
- **Key Influencers Visual:** Attrition Drivers
- **Matrix:** Total income by work shift and job role
- **Clustered Column Chart:** Monthly Income by Age
- **Bar Chart:** Promotion Eligibility by Income Summary

1. HR Overview Dashboard



HR OVERVIEW DASHBOARD

Employee_Count
232

Total_Income
₹ 1.762817...

Average_Income
₹ 7.59834...

High_Performance_Income
₹ 720.419K

Attrition_Count
45

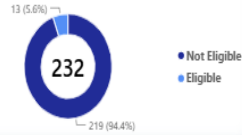
Average_Training_Hours
32.34

Average_Leave
11.75

Ranking

Ranking	Department	Sum of Monthly_Income
1	Human Resources	217967
2	Research & Development	996571
3	Sales	548279
3		1762817

Employee_Count by Promotion_Eligibility



Average_Income



Gender

☒ Select all

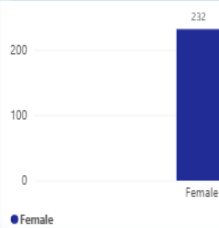
☒ Female

☐ Male

☐ Non-Binary

☐ Unknown

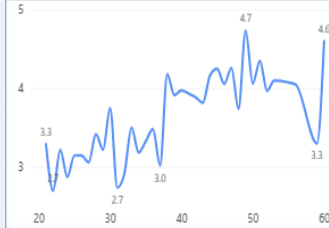
Count of Job_Role by Gender



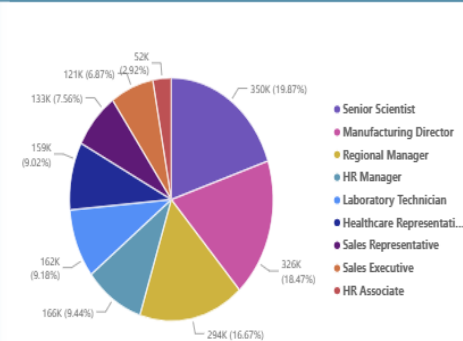
Count of Work_Shift and Sum of Work_Experience by Gender



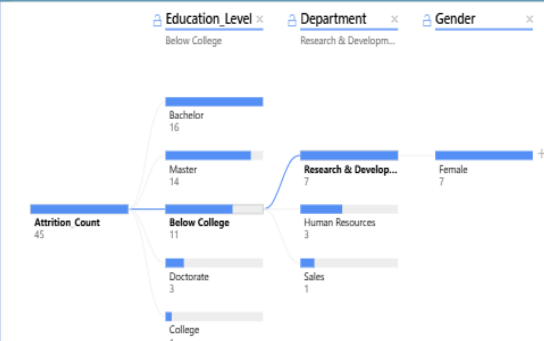
Average of Performance_Score by Age



Sum of Monthly_Income by Job_Role



Decomposition_Tree



2. HR Analysis Dashboard





Key Insights

- Workforce Composition**

The organization maintains a balanced and diverse workforce of 500 employees across multiple departments and gender groups.

- Income & Compensation**

Mid-career employees (ages 40–45) contribute the highest income, showing a strong link between experience and earning potential.

- Performance**

Performance scores steadily increase with age and are highest among employees with Bachelor's and Master's degrees.

- **Promotion Eligibility**
Medium-income employees form the largest group eligible for promotion, indicating a strong internal talent pipeline.
- **Attendance & Work Shift**
Attendance rates vary by education level, highlighting specific groups that may require engagement or policy support.
- **Training**
Training hours are concentrated in operational roles, showing targeted skill development in high-impact job categories.
- **Attrition**
Attrition is highest in Research & Development and is strongly influenced by young age (≤ 21), low experience (≤ 20), and certain job roles.
- **Departmental Insights**
Research & Development contributes the most income but also faces the highest attrition, indicating workload or role-fit challenges.
- **Income by Age**
Income peaks around ages 40–45, confirming that mid-career employees are the strongest financial contributors.
- **Decomposition Tree**
Master's degree holders show the highest attrition, revealing dissatisfaction among highly qualified employees.
- **Data Modelling**
A centralized HR Details table connected to multiple dimension tables ensures accurate filtering, faster performance, and reliable insights across dashboards.

Conclusion

The HR dashboards successfully transform raw employee data into clear, actionable insights that help the organization understand workforce performance, income patterns, attrition risks, and training needs. By visualizing key HR metrics through an efficient data model, the analysis enables informed decision-making, supports strategic planning, and strengthens overall workforce management.